

Experiment No. 05

Polynomial Regression, Overfitting, Ridge and Lasso Regularisation

Aim:

To study polynomial regression and the effect of increasing model complexity on training and testing errors, identify overfitting, and apply **Ridge and Lasso regularisation** to control model complexity and reduce overfitting.

Objective:

After completing this experiment, students will be able to:

1. To understand the concept of **polynomial regression**.
2. To investigate how polynomial degree affects model complexity.
3. To compare **training MSE and testing MSE** for different polynomial degrees.
4. To identify the point at which a model begins to **overfit**.
5. To understand the difference between underfitting and overfitting.
6. To apply **Ridge (L2) regularisation** to a high-degree polynomial model.
7. To apply **Lasso (L1) regularisation** to a high-degree polynomial model.
8. To study how Ridge shrinks model coefficients.
9. To study how Lasso can force some coefficients to exactly zero.
10. To select a suitable polynomial degree and regularisation method based on test performance.

Pseudo Algorithm :

Input: Housing Prices Dataset (“<https://www.kaggle.com/datasets/yasserh/housing-prices-dataset>”)

Steps:

1. **Start**
2. Load the house-pricing dataset.
3. Select one numerical predictor and the target variable.
4. Divide the data into training and testing sets.
5. Predict how train and test errors will change as polynomial degree increases.
6. Create polynomial features for degrees **1, 3 and 10**.
7. Train a linear regression model for each degree.
8. Calculate training MSE and testing MSE.
9. Store the results in a table.
10. Plot training and testing MSE against polynomial degree.
11. Identify the degree at which the model begins to overfit.
12. Select the high-degree polynomial model as the overfit model.
13. Apply **Ridge regularisation** using different values of α .
14. Apply **Lasso regularisation** using the same or comparable values of α .
15. Calculate test MSE for each regularised model.
16. Examine the magnitude of Ridge coefficients.
17. Count the number of zero coefficients produced by Lasso.
18. Compare unregularised, Ridge and Lasso models.
19. Try an intermediate polynomial degree such as 3 or 5 with regularisation.

20. Select the final model based primarily on test performance and model simplicity.
21. Write the final conclusion.
22. **Stop**

Sample Outcome:

- The polynomial regression models were successfully trained for degrees 1, 3, and 10.
- The degree-1 model showed relatively high training and testing errors, indicating that the relationship was not completely captured by a straight line.
- Increasing the degree to 3 reduced the testing error and improved the model's ability to capture the relationship between house area and price.
- At degree 10, the training error became very small while the testing error increased substantially. This indicated overfitting.
- Ridge and Lasso regularisation were then applied to the degree-10 model. Regularisation reduced the effect of large polynomial coefficients and improved the generalisation of the high-degree model.

The numerical results shown below are illustrative. Record the actual values produced by your dataset.

Sample Observation:

- As polynomial degree increases, **training MSE generally decreases**.
- Test MSE initially decreases because the model captures more useful patterns.
- At very high degrees, test MSE can increase because the model starts fitting noise.
- The difference between training and test error is called the **generalisation gap**.
- A large generalisation gap is a warning sign of overfitting.
- Ridge shrinks polynomial coefficients toward zero.
- Lasso can shrink some coefficients **exactly to zero**.
- Increasing α generally increases the strength of regularisation.
- An intermediate degree with suitable regularisation can outperform an unregularised degree-10 model.
- The final model should be chosen based primarily on **test/validation performance**, while also considering model complexity.

Tools / Software Required:

Hardware: Computer/Laptop, Minimum 4 GB RAM

Software: Python 3.x, Jupyter Notebook or Google Colab, Anaconda (optional)

Python Libraries: NumPy, Pandas, Matplotlib, Scikit-learn, Seaborn, Joblib / Pickle

Viva Questions:

1. What is polynomial regression?
2. What happens when the polynomial degree is increased?
3. What is the difference between Ridge and Lasso regression?
4. What is the role of the parameter α ?
5. Why is a degree-10 polynomial more likely to overfit?
6. What is coefficient shrinkage?
7. What is sparsity in Lasso regression?
8. Why should we compare train MSE and test MSE?
9. Why might an intermediate polynomial degree be preferred over degree 10?
10. What is the bias-variance trade-off in polynomial regression?